Try these

1. Express the following exponent numbers using $a^m \times b^m = (a \times b)^m$.

   (i) $5^2 \times 3^2$

   (ii) $x^3 \times y^3$

   (iii) $7^4 \times 8^4$

2. Simplify the following exponent numbers by using $\left(\frac{a}{b}\right)^m = \frac{a^m}{b^m}$.

   (i) $5^3 \div 2^3$

   (ii) $(-2)^4 \div 3^4$

   (iii) $8^6 \div 5^6$

   (iv) $6^3 \div (-7)^3$

Example 3.10 Simplify by using the law of exponents.

(i) $7^6 \times 3^6$ (ii) $4^3 \times 2^3 \times 5^3$ (iii) $72^5 \div 9^5$ (iv) $6^{13} \times 48^{13} \div 12^{13}$

Solution

(i) $7^6 \times 3^6 = (7 \times 3)^6 = 21^6$ [Since, $a^m \times b^m = (a \times b)^m$]

(ii) $4^3 \times 2^3 \times 5^3 = (4 \times 2 \times 5)^3 = 40^3$ [Rule extended for 3 numbers]

(iii) $72^5 \div 9^5 = (72 \div 9)^5 = 8^5$ [Since, $\frac{a^m}{b^m} = \left(\frac{a}{b}\right)^m$]

(iv) $6^{13} \times 48^{13} \div 12^{13} = 6^{13} \times (48^{13} \div 12^{13})$ [BIDMAS]

$= 6^{13} \times \left(\frac{48}{12}\right)^{13}$ [Since, $\frac{a^m}{b^m} = \left(\frac{a}{b}\right)^m$]

$= 6^{13} \times 4^{13}$

$= (6 \times 4)^{13}$ [Since, $a^m \times b^m = (a \times b)^m$]

$= (24)^{13}$

G5T5T2

DO YOU KNOW?

1. All the 10 digits appear once in the expansion of 32043^2. That is, the value of $32043^2 = 1026753849$.
2. There are beautiful equations with same exponent consecutive natural numbers as the base.

$$3^2 + 4^2 = 5^2$$
$$10^2 + 11^2 + 12^2 = 13^2 + 14^2$$

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